

■■ Installation & Setup Guide

Step-by-step instructions to set up and run the project

Prerequisites

- A computer with Windows / macOS / Linux
- USB cable (Micro-USB for NodeMCU)
- All hardware components assembled and wired correctly
- Arduino IDE installed (version 1.8.x or 2.x)
- Internet connection (for installing ESP8266 board package)

Step-by-Step Setup

Step 1	Install Arduino IDE Download Arduino IDE from https://www.arduino.cc/en/software Install and launch the application.
Step 2	Add ESP8266 Board Support Go to: File > Preferences In 'Additional Board Manager URLs', paste: https://arduino.esp8266.com/stable/package_esp8266com_index.json Click OK. Then: Tools > Board > Board Manager Search 'esp8266' and install 'esp8266 by ESP8266 Community'.
Step 3	Install CH340 USB Driver NodeMCU V3 uses CH340 USB-to-Serial chip. Download and install the CH340 driver for your OS: Windows: ch341ser.exe macOS: CH34xVCPDriver Linux: built-in
Step 4	Clone the Repository Open terminal / command prompt and run: <pre>git clone https://github.com/YOUR_USERNAME/automatic-exhaust-fan.git cd automatic-exhaust-fan</pre>
Step 5	Open the Sketch Open Arduino IDE File > Open > navigate to src/automatic_exhaust_fan.ino The sketch will open in the editor.
Step 6	Select Board & Port Tools > Board > ESP8266 Boards > NodeMCU 1.0 (ESP-12E Module) Tools > Upload Speed > 115200 Tools > Port > Select the COM port (Windows: COMx, Mac/Linux: /dev/ttyUSB0)

Step 7	Upload the Code Click the Upload button (right arrow →) or press Ctrl+U Wait for 'Done uploading' message in the status bar. The NodeMCU will restart automatically.
Step 8	Test the System Open: Tools > Serial Monitor Set baud rate to 9600 You should see 'Smoke Level: XXX' printing every second. Bring a lighter (unlit, just gas) or smoke near the MQ-2 sensor. The fan, buzzer, and red LED should activate when smoke > 500.

Troubleshooting

Problem	Solution
Port not found	Install CH340 driver. Try a different USB cable (data cable, not charge-only).
Upload fails	Hold FLASH button on NodeMCU while clicking Upload, release after upload starts.
Sensor always reads high	Allow 20 seconds warm-up time. Clean the sensor if dusty.
Fan not turning on	Check relay wiring. Verify relay LED blinks when D4 is triggered.
Serial monitor shows nothing	Check baud rate is set to 9600 in Serial Monitor.